Institute of Technology of Cambodia

Department of Industrial and Mechanical Engineering

Subject: Introduction to Industrial Engineering
Case Study: Health Care

Topic: Surgery for Cancer Treatment

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Surgery for Cancer Treatment

I. Introduction

Cancer is a broad group of diseases characterized by the uncontrolled growth and spread of abnormal cells in the body. Normally, the body's cells grow and divide in a controlled manner, but when this process goes awry, it can lead to the formation of a tumor or the spread of malignant cells throughout the body. This uncontrolled growth can occur in any part of the body, and cancer can affect nearly every organ and tissue. Moreover, Cancer is a serious disease that has been faced for a long time. In the past, treatment was done using traditional methods such as using natural remedies, acupuncture, or, in some cases, cutting the flesh. Knowledge about the disease was unclear, and technology was not yet developed, so the treatment was ineffective. However, in the present era, the development of science and technology has led to modern treatment methods such as surgery, chemotherapy, radiation therapy, immunotherapy, and targeted therapy. Psychological support and nutritional care are also provided. Thanks to modern technology, patients can now receive highly effective treatment and have a longer chance of survival.

1. Problem

Curing cancer remains a significant challenge due to the complexity of the disease, variability in patient responses to treatment, and limitations in current medical technologies. Moreover, cancer is different from other diseases, since it doesn't show about signs that the patient can know it is a cancer. It has many levels and many types, that why it is hard to define and find the best ways to treat and kill cancer.. Despite advancements in research and therapies, achieving complete remission remains difficult, leading to prolonged treatment durations, high healthcare and emotional distress for patients and their families.

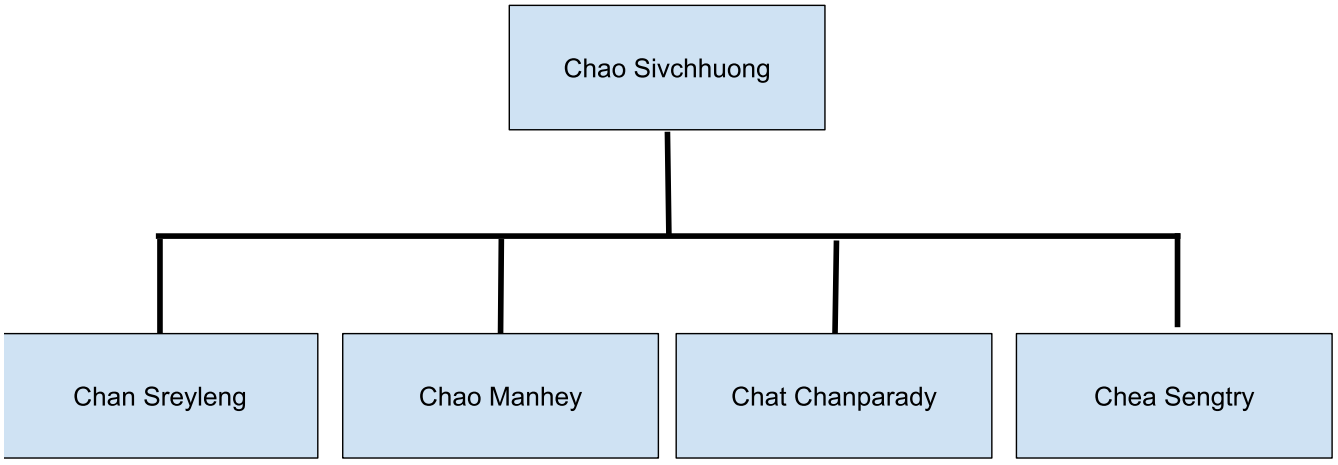
2. Objective

- Improve Access to Surgery
- Minimize Patient Waiting Time
- Enhance Patient Comfort & Recovery
- Sustainability in Healthcare

3. Scope of work

3.1. Team Members

Team Members



Responsibility of team members

Chan Sreyleng	Chao Manhey	Chao Sivchhuong	Chat Chenparady	Chea Sengtry
Responded to **-Introduction** **-Review** **-Slide**	**Responded to** **-Problem** **-Objective** **-Scope of work**	**Responded to** **-Methodology** **-Check** **-Discuss**	**Responded to** **-Result** **-Conclusion** **-Create book**	**Responded to** **-Problem** **-Objective** **-Scope of work**

3.2. Systems Model Approach:

Stage 0 Cancer (Carcinoma in situ / Pre-cancerous)

- Inputs:
  - Genetic mutations
  - Environmental triggers (smoking, UV, toxins)
  - Family history
  - Early cell changes detected via screening
- Processes:
  - Cellular dysplasia without invasion
  - Localized abnormal cell growth
  - No spread to other tissues
- Outputs:
  - Detection via screening (e.g., Pap smear, mammogram)
  - Diagnosis as Stage 0
  - Often asymptomatic
- Feedback Loops:
  - Continued surveillance
  - Adjust lifestyle or environment to reduce risks
  - Regular re-screening
- Control Mechanisms:
  - Surgical removal (e.g., excision or localized procedure)
  - Monitoring (watchful waiting)
  - Chemoprevention in some cases

Stage I Cancer

- Inputs:
  - Localized tumor growth
  - Continued mutation and unchecked cell division
  - Early symptoms or incidental detection
- Processes:
  - The tumor remains within the original tissue
  - No lymphatic or vascular invasion
- Outputs:
  - Diagnosis from imaging or biopsy
  - Prognosis generally favorable
  - Potential surgical resection

- Feedback Loops:
    - Pathology results determine margins and next steps
    - Ongoing monitoring for recurrence
- Control Mechanisms:
    - Surgery
    - Localized radiation
    - Adjuvant therapy if indicated

Stage II Cancer

- Inputs:
    - Larger tumor size
    - Possible local lymph node involvement
    - Symptom development (e.g., lump, bleeding)
- Processes:
    - Tumor invades deeper tissues
    - Begins spreading via the lymphatic system
- Outputs:
    - Imaging shows tumor size and spread
    - Biopsy confirms staging
    - Potential increase in symptoms
- Feedback Loops:
    - Treatment response monitored
    - Biomarkers tracked
    - Adjust treatment plans based on progression
- Control Mechanisms:
    - Surgery with wider margins
    - Chemotherapy and/or radiation
    - Regular follow-ups and scans

Stage III Cancer

- Inputs:
    - Deep tissue invasion
    - Spread to multiple lymph nodes
    - Significant tumor burden
- Processes:
    - Tumor impacts surrounding organs/tissues
    - Cancer becomes harder to fully remove

    - Higher risk of recurrence
- Outputs:
    - Diagnosis of advanced local disease
    - Complex treatment regimen
    - Prognosis depends on tumor type and response
- Feedback Loops:
    - Tumor markers and scans guide treatment efficacy
    - Treatment side effects influence the approach
    - Multi-disciplinary team adjusts care dynamically
- Control Mechanisms:
    - Neoadjuvant (pre-surgery) chemo/radiation
    - Surgery (if resectable)
    - Aggressive systemic therapy
    - Immunotherapy/targeted therapy in some cases

Stage IV Cancer (Metastatic)

- Inputs:
    - Cancer cells spread via blood or lymph
    - Secondary tumors in distant organs
    - System-wide symptoms (e.g., fatigue, pain, weight loss)
- Processes:
    - Metastasis to liver, lungs, bones, brain, etc.
    - Systemic impact on organ function
    - Cancer becomes chronic/terminal in many cases
- Outputs:
    - Diagnosis through PET, CT, MRI, and biopsy
    - Variable survival depending on cancer type and spread
    - Reduced quality of life if unmanaged
- Feedback Loops:
    - Treatment effectiveness drives ongoing changes
    - Palliative care involvement
    - Regular reassessment of goals (quality vs. quantity of life)
- Control Mechanisms:
    - Systemic chemotherapy
    - Targeted therapies
    - Palliative care to manage symptoms
    - Hospice in late-stage cases

3.3. Points upgrade

a. Improving Access to Surgery

Tasks & Strategies:

- Increase the number of surgical units in public hospitals.
- Implement mobile surgical teams for remote areas.
- Offer specialized training programs for surgeons.
- Introduce telemedicine for pre-surgical consultations.

Supporting Data & Case Studies:

- WHO reports indicate that **5 billion people lack access to safe surgery** globally.

Systems Model Approach:

- **Stock & Flow Diagram**: Mapping patient inflow and available surgical capacity.
- **Causal Loop Diagram**: Demonstrating how increased training impacts accessibility.

c. Minimizing Patient Waiting Time

Tasks & Strategies:

- Implement digital scheduling and tracking systems.
- Increase the number of available operating rooms.
- Streamline administrative procedures for faster approvals.
- Expand staff availability for pre-op and post-op care.

Supporting Data & Case Studies: Delays of over 3 months in cancer surgery increase

- Studies show that **mortality rates increase by 20%**.
- **Using digital scheduling, Canada's "Wait Time Alliance" reduced surgical delays by 50%**.

Systems Model Approach:

- **Process Flowchart:** Mapping the patient's journey from diagnosis to surgery.
- **Bottleneck Analysis:** Identifying points of delay in the surgical process.

d. Enhancing Patient Comfort & Recovery

Tasks & Strategies:

- Enhance post-surgical pain management and psychological support.
- Improve hospital environments for patient recovery.
- Develop rehabilitation and physical therapy programs.
- Train staff in holistic patient care.

Supporting Data & Case Studies:

- **Harvard Medical School research** shows that better post-op care reduces hospital readmission by **30%**.
- **Japan's hospitals** use robotic-assisted rehabilitation to speed up patient recovery.

Systems Model Approach:

- **Stock & Flow Diagram:** Showing patient recovery rates and medical staff availability.
- **Causal Loops:** Linking patient comfort measures with faster recovery times.

e. Sustainability in Healthcare

Tasks & Strategies:

- Reduce surgical waste through sustainable disposal practices.
- Implement energy-efficient hospital infrastructure.
- Promote the use of reusable surgical materials where possible.
- Develop hospital sustainability policies.

Supporting Data & Case Studies:

- **Healthcare contributes to 4.4% of global carbon emissions** (The Lancet, 2022).
- **Case Study:** The UK's NHS reduced hospital waste by **60%** through eco-friendly disposal programs.

Systems Model Approach:

- **Sustainability Feedback Loop:** Showing how eco-friendly policies impact costs and environmental health.
- **Life Cycle Analysis:** Evaluating the environmental impact of surgical resources.

II. Review

Over the years, Cancer Healthcare systems have started looking for ways to improve their services. The most important is in cancer care, Six Sigma can be used to reduce variability in treatment protocols, minimize medication errors during chemotherapy, and improve the accuracy of radiation therapy delivery.

Analyzing data on treatment outcomes, complications, and patient satisfaction can help identify areas for improvement and implement evidence-based solutions. Cancer patients often face long waiting times for appointments, treatments, and diagnostic procedures. Reducing waiting times is critical in cancer care, as delays can impact treatment effectiveness and patient well-being.

Optimizing Healthcare Processes:

Industrial engineering techniques can be used to optimize the entire cancer care pathway, from diagnosis to survivorship. In summary, while progress has been made in these areas, more research is needed to understand the full impact of process optimization, resource management, and technology on cancer healthcare systems.

III. Methodology

We can use methodologies like **LEAN**, **SIX SIGMA**, and **ERGOLOGY** to estimate and evaluate the impact of process optimization, waste time, cost reduction, and improved efficiency in surgery for cancer treatment. These are beneficial and powerful methods in many sectors, such as companies, hospitals, etc.

1. **Improving the quality of surgery for cancer treatment**
   - **Six Sigma Methodology**

What is Six Sigma?

Six Sigma is a set of techniques and tools used to improve business processes. It was introduced in 1986 by engineer Bill Smith while working at Motorola. Six Sigma practitioners use statistics, financial analysis, and project management to identify and reduce defects and errors, minimize variation, and increase quality and efficiency.

The Six Sigma is

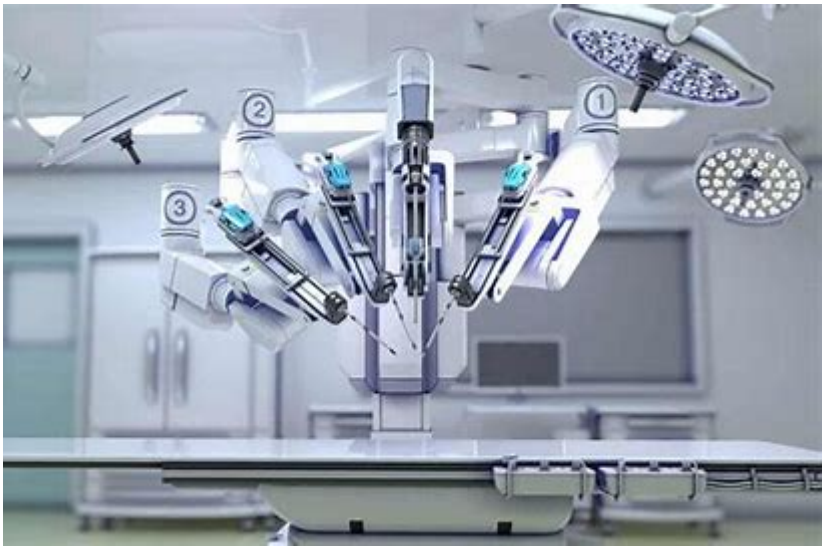
1. Define
2. Measure
3. Analyze
4. Improve
5. Control
6. Verify

Using **Six Sigma Methodology**, hospitals and clinics can improve their quality and efficiency and reduce medical errors in cancer surgery and medication.

1.1. Using the appropriate equipment

Ensuring that surgical procedures are performed with modern, high-quality medical equipment to enhance precision, reduce complications, and improve patient outcomes. This includes advanced imaging systems, robotic-assisted surgery tools, and minimally invasive instruments.

Robotic-assisted surgery



Using all these ways to treat the level of cancer:

1. X-ray

Used for:

- Detecting lung cancer, bone tumors

Limitations:

- Cannot detect small or early-stage
- tumors limited to soft tissues

2. CT Scan (Computed Tomography)

Used for:

- Detecting tumors in the **lungs, liver, pancreas, and bones**
- Identifying cancer spread (metastasis)
- Planning radiation therapy

Limitations:

- Uses radiation
- Less detailed for soft tissues compared to to MRI

3. MRI (Magnetic Resonance Imaging)

Used for:

- **Brain, spinal cord, breast, and soft tissue tumors**
- Detecting the size and extent of cancer
- Differentiating between benign and malignant tumors

Limitations:

- Expensive
- Slower scan time

4. PET Scan (Positron Emission Tomography)

Used for:

- **Detecting active cancer cells** (metabolic activity)
- Checking if cancer has spread (metastasis)
- Evaluating response to treatment

Limitations:

- Requires injection of a radioactive
- tracer Not always specific (some infections or inflammation can appear as cancer)

5. SPECT Scan (Single-Photon Emission Computed Tomography)

Used for:

- **Bone cancer detection**
- Finding metastases in the bones

Limitations:

- Less commonly used compared to PET

6. Ultrasound (Sonography)

Used for:

- Detecting tumors in the liver, breast, thyroid, and reproductive organs
- Guiding biopsies

Limitations:

- Cannot detect cancer spread
- Less effective for deep tissues

So, for easier understanding, we will give you a table summarizing each machine's function.

Best Choices for Different Cancers

Cancer Type	Best Imaging Method
Lung Cancer	CT scan, PET scan, X-ray (early detection)
Brain & Spinal Cancer	MRI, PET scan
Breast Cancer	MRI, Ultrasound, Mammogram (X-ray)
Liver & Pancreatic Cancer	CT scan, MRI, Ultrasound
Bone Cancer	X-ray, CT scan, PET scan, SPECT
Lymphoma (Lymph Nodes)	PET scan, CT scan
Prostate Cancer	MRI, Ultrasound

Using all these ways to know the level of cancer:

1. Imaging Tests (For Size & Spread)

These tests help doctors see how far the cancer has spread:

- CT Scan / MRI: This checks if the tumor has invaded nearby tissues or organs.
- PET Scan: Detects cancer spread (metastasis) in the body.
- X-ray / Ultrasound: Helps find tumors but does not confirm the stage.

2. Biopsy (For Cancer Type & Grade)

- A small tissue sample is taken from the tumor and examined under a microscope.
- It confirms if the tumor is **benign (non-cancerous)** or **malignant (cancerous)** and determines the **grade** (how aggressive the cancer is).

3. Blood Tests (For Cancer Markers)

- Some cancers release specific substances into the blood called **tumor markers** (e.g., PSA for prostate cancer, CA-125 for ovarian cancer, AFP for liver cancer).

- Helps doctors track cancer progression but **cannot** confirm staging alone.

4. Bone Marrow Tests (For Blood Cancers)

- If cancer, like **leukemia or lymphoma,** is suspected, a **bone marrow biopsy** can show how far it has spread in the blood.

1.2. Increase Training & Skill Development

Providing continuous education and hands-on training for surgeons and medical staff to enhance their expertise in cancer surgery. This includes workshops, simulation-based learning, and international collaborations to stay updated with the latest advancements.

1. Medical & Nursing Students

- Hospitals & Cancer Centers – Medical and nursing students can do clinical rotations in oncology departments.
- Public Health Programs – Students interested in cancer prevention & community health can work with NGOs or government projects.

Example:

- Calmette Hospital (Cambodia) allows medical students to do oncology training.
- The National Cancer Institute (NCI, USA) offers medical fellowships.

2. Engineers & Scientists

- Biomedical Engineers help design medical devices (MRI, CT, radiation therapy machines).
- Mechanical Engineers Work on ergonomic hospital equipment and radiation shielding systems.
- Pharmacists & Chemists – Study chemotherapy drugs and cancer treatments.

Example:

- MIT & Harvard Medical School collaborate on cancer research with engineering students.
- The Institute of Technology of Cambodia (ITC) could connect with hospitals for medical device research.

3. Public Health & Systems Thinking Students

- Cancer Prevention & Screening Programs – Students help design policies

to improve public healthcare.
- Data Analysis & AI in Healthcare – Engineering and IT students work on AI-based cancer detection.
- Ergonomics Research – Studies how hospital design and equipment improve patient comfort.

Example:

- WHO's Internship Program includes cancer research projects.
- Local NGOs in Cambodia work on cancer awareness and screening campaigns.

4. Internships, Fellowships & Volunteering

- Medical Internships – Hands-on experience in oncology hospitals.
- Engineering & Research Fellowships – Work on radiation therapy technology and medical devices.
- Volunteering with Cancer NGOs – Help with patient care, fundraising, and awareness.

Example:

- UICC (Union for International Cancer Control (UICC) offers global cancer research internships.
- Sihanouk Hospital Center of HOPE (Cambodia) has programs for medical trainees.

We must sign an MCU with these universities and these communities to exchange and improve our staff's skills in cancer.

1.3. Ensuring Post-Surgical Monitoring & Follow-up Care

Establishing a structured follow-up system to monitor patients after surgery, ensuring timely interventions if complications arise. This includes remote monitoring, post-discharge check-ups, and personalized rehabilitation programs to improve long-term outcomes.

- Summary: Best Practices for Post-Surgical Cancer Monitoring

Phase	Key Actions
First 48 Hours	Pain control, wound care, early mobility, vital sign monitoring
First 3 Months	Regular checkups, blood tests, imaging, and physical therapy
6 Months - 5 Years	Cancer screenings, lifestyle modifications, and long-term side effect management
Home Monitoring	Telemedicine, remote patient monitoring, virtual doctor consultations
Support System	Family education, cancer support groups, financial & mental health assistance

2. Reducing waiting in surgery for cancer treatment

- **Lean Methodology**

What is Lean?

Lean is a systematic approach to continuous process improvement through the elimination of waste. It originated in the manufacturing industry, specifically within Toyota, in the mid-20th century as part of the Toyota Production System (TPS), which sought to maximize value for the customer through the optimal use of resources. 8 Waste of Lean:

1. Defects
2. Overproduction
3. Waiting
4. Non-Utilized Talent
5. Transportation
6. Inventory
7. Motion

Using the Lean Methodology, hospitals and clinics can improve workflow, shorten patient waiting times, and improve discharge.

2.1. Quick appointment

Implementing a streamlined scheduling system to ensure that cancer patients receive timely surgical appointments. This includes prioritizing urgent cases, reducing administrative delays, and using automated booking systems.

2.1.1. Finding an Oncologist or Cancer Center 🏥

To ensure a quick appointment for cancer treatment, consider visiting specialized cancer treatment centers or hospitals that focus on oncology. These

centers usually have more resources and faster processing for cancer cases. Here are a few steps to help find an oncologist or cancer center:

- Search for Specialized Centers: Major cancer treatment centers often have dedicated teams for different cancer types and can process your case faster. You can look for well-known centers like:
    - Memorial Sloan Kettering Cancer Center (USA)
    - MD Anderson Cancer Center (USA)
    - Royal Marsden Cancer Center (UK)
    - Sihanouk Hospital Center of HOPE (Cambodia)
    - Calmette Hospital (Cambodia)
- Online Booking Systems: Some hospitals and clinics allow you to schedule appointments directly online. This is especially helpful to expedite your booking. Call or visit their websites for availability and details.
- Contact Your Insurance or Local Health Networks: If you are covered by insurance, they may have a list of preferred cancer treatment centers and can help you get an urgent referral or appointment.

2.1.2. Priority Appointments & Urgent Care Services

If you're experiencing more advanced cancer symptoms or have a confirmed diagnosis, you may be eligible for priority appointments. Many specialized cancer centers have mechanisms to help expedite the treatment process:

- Request an Urgent Referral: If you already have a diagnosis or have been referred by your primary care doctor, contact the hospital and request that your case be marked as urgent. Many hospitals will prioritize cancer patients based on the urgency of the situation.
- Emergency Oncology Departments: Some cancer centers have dedicated emergency departments for cancer patients where you can receive immediate consultation. These centers are equipped to handle urgent cancer diagnoses and provide swift treatment options.
- Fast-Track Programs: Some hospitals have "fast-track" pathways where patients with a high suspicion of cancer or those in the later stages are given rapid access to diagnostic and treatment resources.

2.1.3. Support for Rapid Diagnosis in Cambodia

For rapid cancer diagnosis and treatment in Cambodia, here are some recommended steps and locations:

- Calmette Hospital: This is one of the primary government hospitals with

cancer treatment services. They offer cancer surgeries, chemotherapy, and radiation. You can contact them directly for a quick appointment or visit their oncology department.

- Sihanouk Hospital Center of HOPE: This hospital is known for providing affordable cancer care and has a specialized oncology team. They offer chemotherapy, surgical treatments, and follow-up care.
    - They also have a quick referral system for cancer patients and offer support for financial assistance.
- Khmer-Soviet Friendship Hospital: Another prominent hospital in Cambodia, known for its oncology department, may offer quicker consultations and treatment options for cancer patients.
- Contacting Local Cancer Foundations: NGOs and local cancer support foundations like the Cambodia Cancer Society might assist in speeding up the process, offering support for both medical care and financial issues.

2.2. Efficient Resource Allocation

Optimizing the use of hospital resources such as operating rooms, medical staff, and equipment to maximize efficiency. This involves strategic scheduling, reducing idle time in surgery rooms, and balancing patient load across different facilities.

1. Understand the System (Systems Thinking Approach)

Before allocating resources, map out the cancer care system. This includes:

- **Inputs**: Budget, medical staff, facilities, equipment, drugs

- **Processes**: Screening, diagnosis, treatment, follow-up
- **Outputs**: Patient outcomes, survival rates, quality of life
- **Stakeholders**: Patients, doctors, policy makers, suppliers, insurers

A **Causal Loop Diagram (CLD)** or **Stock and Flow Diagram** can help visualize this.

2. Prioritize Based on Evidence & Data

Use **evidence-based medicine** and **epidemiological data** to guide decision-making:

- Focus on **early detection** (often more cost-effective than late-stage treatment).
- Allocate more to cancers with **higher prevalence** and **better treatment response**.
- Use **Health Technology Assessment (HTA)** to evaluate the cost-effectiveness of treatments (like chemotherapy, radiation, immunotherapy).

3. Optimize the Use of Expensive Treatments

Some treatments, like immunotherapy, are very expensive. Efficient allocation includes:

- Using **biomarkers** to identify patients who are most likely to benefit.
- Applying **clinical guidelines** to avoid unnecessary or ineffective treatments.
- Negotiating **bulk pricing or generics** for essential cancer drugs.

4. Decentralize When Possible

- Set up **regional cancer centers** to reduce travel time and overcrowding in central hospitals.
- Train **local healthcare workers** to provide basic screening and follow-up care.

5. Collaborate with Other Sectors

- Work with NGOs, the private sector, and international donors to fund or supply specific areas (like mammogram units, HPV vaccines).
- Leverage **public-private partnerships (PPPs)** to co-manage services.

6. Monitor and Adjust

- Create **KPIs** (Key Performance Indicators) like treatment completion rate, 5-year survival rate, etc.
- Use **digital tools** to track resource use in real time.
- Reallocate based on outcomes — continuously improve.

7. Ensure Equity

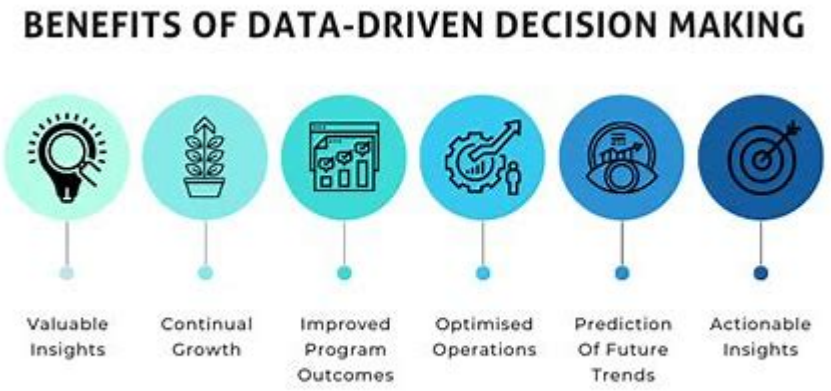
- Use **resource allocation models** that account for social vulnerability (poor, rural, minorities).
- Apply **triage protocols** in a way that doesn't disadvantage marginalized groups.

8. Tools and Methods You Can Use

- **Cost-effectiveness analysis (CEA)**
- **Discrete event simulation** to model hospital flows
- **Linear programming** for optimal allocation
- **Multi-criteria decision analysis (MCDA)**

2.3. Data-Driven Decision-Making

Utilizing patient data analytics and hospital performance metrics to improve surgical workflow and reduce wait times. Predictive modeling can help anticipate patient demand, optimize staff allocation, and identify bottlenecks in the surgical process.



Making a treatment decision

Choosing the best treatment for your situation is a decision that needs to be made after all information has been shared with you and after you've had time to ask questions and have them answered. This process is called informed consent16 and allows people to play an active role in making decisions that affect their health. Some types of cancer have a very limited number of treatment options, so deciding which is best might not be difficult. It's also possible that treatment needs to be started quickly, so there might not be as much time to decide as you'd 5 American Cancer Society cancer.org like. But, in many cases, it can be helpful to get a second opinion before making your decision. Don't be afraid to ask if there's time to get one, and remember you have the right to a second opinion. You can learn more in Seeking a Second Opinion 17. Shared decision-making is a process where decisions about your care are made with you rather than someone making the decisions for you. Shared decision-making involves doctors and other members of the patient's health care team working with you and your family or caregivers to make decisions together. The process involves sharing information back and forth.

Shared decision-making

It can be hard in some cases when there are many doctors, specialists, or caregivers involved in helping a patient make decisions. It's not always easy to get everyone's input, and it can take some work and reminding to keep information flowing well. But shared decision-making can help make sure you get the right tests and treatment and that the right type of care is planned around them. It considers risks, benefits, and costs of treatment, along with any other concerns you might have.

If you're not sure you want to get treatment

Not everyone diagnosed with cancer decides to get treatment or to be treated right away. That is always a choice, but make sure you understand why you are thinking about not getting treatment. It's normal to feel anxious and

overwhelmed by the amount of information you've been given. Make sure you get answers to all the questions you have to understand what is likely to happen if you do and if you do not get treatment for your cancer. Some people have heard that cancer treatment is worse than the cancer itself. Getting through cancer treatment can be tough, but not getting treatment lets some types of cancer grow unchecked. Depending on the type, an untreated cancer can lead to new and worse symptoms and can be fatal. Also, getting treatment soon after diagnosis usually works better for most types of cancer than waiting until the cancer is worse, when treatment probably won't work as well. If you or a loved one is thinking of refusing or delaying cancer treatment, talk with your doctor or nurse. If you're not getting the answers you need, you might think about getting a second opinion 18. It might also help to talk to someone else who has been treated for your type of cancer. Look for local or online support groups19 or join a cancer survivor community20 to connect with others. 6 American Cancer Society cancer.org

When was the decision made?

Even when you have learned everything you can about your treatment options and you've made your decision, it's important to keep communicating with your cancer care team as questions come up. Here are some things to think about: Be sure to ask any specific questions about the treatment you'll be getting before it starts. Even if you've asked the same questions before, you might feel better getting the answers again. If you don't understand something, ask your cancer care team to explain it better. If you forget something, ask about it again. Bring up questions at every treatment visit as you think of them. Plan for side effects. Each type of treatment has its own set of side effects that can be expected. It's important to know about them, how to manage them, when to tell someone about them, and when to get help for them. It's a good idea to ask your cancer care team how information will be shared with other members of your health care team, such as your surgeon, radiation oncologist, family doctor, and any specialists that might be involved in your care. Find out the best way to get in touch with your cancer care team. You should know the phone numbers to call during weekdays and on weekends. Ask about how to call for any emergencies, too. There also might be an online patient portal you can get into to send messages and look up information about your care.

3. Optimize healthcare spaces to support both caregivers and patients

- **Ergonomics Methodology**

What is Ergonomics?

Ergonomics is the study of matching job requirements and the environment to the worker to maximize efficiency, quality, and quantity of work while minimizing work-related musculoskeletal disorders (WMSD), fatigue, and overexertion.

Ergonomics methodology can be applied in hospitals to improve safety, efficiency, and comfort for patients and healthcare workers. It helps reduce injuries, fatigue, and errors while enhancing patient care quality.

3.1. Patient Comfort

The design of the work environment, particularly in cancer treatment settings, plays a significant role in healthcare worker safety and efficiency. Long shifts and physically demanding tasks can lead to injury, fatigue, and burnout.

- Ergonomic Workspaces: Implement ergonomic furniture and equipment in patient rooms, clinics, and treatment areas to minimize physical strain. Adjustable chairs, workstations, and equipment can reduce the risk of musculoskeletal injuries among healthcare workers.
- Equipment and Tool Accessibility: Ensure that necessary tools and equipment (e.g., chemotherapy infusion pumps, patient monitoring devices, etc.) are easily accessible to avoid overexertion and repetitive strain injuries.
- Workload Balance: Create a balanced workload and allow adequate break times to prevent fatigue, especially for staff who work long shifts. Fatigue can reduce efficiency and increase the risk of mistakes, which can be dangerous for both healthcare workers and patients.
- Safe Patient Handling: Provide training on safe lifting techniques and implement assistive devices (e.g., hoists, lifting slings) to prevent injuries related to lifting or moving patients.

3.2. Healthcare Worker Safety & Efficiency

Training healthcare workers to recognize and mitigate risks is a critical first step to ensuring their safety. Continuous training helps them stay updated on safety protocols, infection control procedures, and emergency response measures.

- Onboarding and Ongoing Training: Provide comprehensive safety training for new staff and continue to offer refresher courses regularly. This should include the use of protective equipment, safe handling of hazardous drugs, infection control, and emergency procedures.
- Infection Control: Train workers on infection prevention protocols such as proper hand hygiene, disinfection procedures, and safe handling of equipment, particularly when working with immunocompromised cancer patients.
- Personal Protective Equipment (PPE): Ensure that all healthcare workers have access to and are trained on the use of necessary PPE, including gloves, gowns, face shields, and masks, especially when handling chemotherapy agents or patients with contagious diseases.

- Hazardous Drug Handling: Chemotherapy drugs are highly toxic and require special protocols. Provide staff with specific training on safe handling, administration, and disposal of hazardous substances, following guidelines such as those from NIOSH (National Institute for Occupational Safety and Health).

3.3. Mental Health & Psychological Support

Healthcare workers in cancer care are often under immense emotional and psychological pressure, which can lead to burnout and compassion fatigue. Creating a supportive, healthy work culture is essential for their well-being and efficiency.

- Mental Health Support: Offer mental health resources for healthcare workers, such as counseling services or support groups. Encourage employees to seek help when dealing with the emotional toll of cancer care.
- Burnout Prevention: Recognize the signs of burnout and create an environment where workers can take mental health breaks, use vacation time, or participate in stress-reducing activities (e.g., yoga or mindfulness).
- Peer Support Programs: Implement peer support systems where healthcare workers can debrief and discuss emotional challenges in a safe and confidential setting.
- Workplace Recognition: Regularly recognize and appreciate healthcare workers for their efforts through awards, thank-you events, or simple gestures of gratitude. This can improve morale and motivate staff.

3.4. Rehabilitation & Post-Treatment Care

Efficient staffing and scheduling are vital for minimizing stress and improving the work-life balance of healthcare workers. An optimized staffing system ensures adequate coverage without overburdening any one individual, helping prevent errors and burnout.

- Flexible Scheduling: Provide flexible scheduling options, such as short shifts, shift rotations, and time-off policies, to give workers adequate rest and balance their personal lives with work.
- Staffing Plans Based on Demand: Implement systems to ensure that the hospital is adequately staffed during peak hours or in high-demand areas (e.g., oncology clinics or chemotherapy treatment rooms).
- Team Collaboration: Design schedules that allow for better team collaboration and ensure that workers have sufficient time to interact and share responsibilities, reducing the likelihood of mistakes and improving

patient care.

- Real-Time Staffing Adjustments: Use staffing software that can adjust schedules dynamically based on patient volume or emergencies. This allows healthcare providers to ensure there are always enough

3.5. Ongoing Evaluation & Feedback Loop

To maintain healthcare worker safety and efficiency, ongoing evaluation and feedback are necessary. Regular assessments ensure that systems, workflows, and safety measures are constantly improving.

- Regular Safety Audits: Conduct safety audits to identify potential hazards, gaps in training, or inefficiencies in workflows that could compromise staff safety.
- Feedback Surveys: Regularly collect feedback from healthcare workers through surveys or focus groups to identify areas where safety can be improved and where workflow efficiency can be enhanced.
- Quality Improvement Initiatives: Create a continuous quality improvement cycle to address issues promptly and improve processes, focusing on worker safety, patient care, and overall efficiency.

IV. Result

Application within Cancer Care:

- **Six Sigma:**
  - Reducing medication errors in chemotherapy preparation and administration.
  - Improving the accuracy and consistency of radiation therapy planning.
  - Minimizing variations in diagnostic testing and pathology reporting.
- **Lean:**
  - Streamlining patient flow in oncology clinics and treatment centers.
  - Reducing waiting times for consultations, imaging, and radiation therapy.
  - Optimizing the supply chain for chemotherapy drugs and other essential materials.
- **Ergonomics:**
  - Designing comfortable and supportive treatment chairs and beds.
  - Optimizing positioning during procedures to minimize discomfort and pain.
  - Creating environments that minimize sensory overload (e.g., controlled lighting, noise levels).

By using data collected from Six Sigma and lean processes, healthcare systems can optimize how they treat patients. This can result in better patient outcomes and reduce the system's cost.

V. Conclusion

Integrating Six Sigma, Lean, and ergonomic principles into cancer care represents a paradigm shift towards a more efficient, patient-centered, and high-quality healthcare system. By leveraging these industrial engineering methodologies, healthcare providers can address the unique challenges of cancer care, reduce variability, eliminate waste, and create a safer and more comfortable environment for patients and staff. The result is improved patient outcomes, reduced costs, and a more sustainable healthcare system. The ongoing application of these engineering principles will be crucial in improving cancer care, ultimately leading to a better quality of life for cancer patients.

VI. References

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